

Patient Assessment: An Integral Part of Pharmacy Practice

March 2025 version



Learning Objectives

- ✓ Gather necessary information from the patient or their advocate to complete assessments
- ✓ Perform comprehensive and reliable patient assessments using assessment frameworks
- ✓ Use the Indication, Efficacy, Safety, Utilization (IESU) model to conduct a therapeutic review of new and refill prescriptions
- ✓ Construct reliable and effective follow-up and monitoring plans
- ✓ Develop comprehensive, yet efficient, documentation methods for recording patient interactions and care plans

Table of Contents

01

Patient Care Process

02

Step 1: Meet the Patient

03

Step 2: Collect Patient Information

04

Step 3: Assess the Patient

05

Step 4: Develop a Care Plan

06

Step 5: Follow-up and Monitoring

07

Documentation

08

Patient Case

09

Putting It All Into Practice

Legend

BID = Two Times Daily

CABG = Coronary Artery Bypass Graft

CCC = Chat, Check, Chart

COPD = Chronic Obstructive Pulmonary Disease

DRP = Drug-Related Problem

IESU = Indication, Efficacy, Safety, Utilization

INR = International Normalized Ratio

NAPRA = National Association of Pharmacy Regulatory Authorities

PDE-5 Inhibitor = Phosphodiesterase-5 Inhibitor

PRN = As Needed

QID = Four Times Daily

SCr = Serum Creatinine

TID = Three Times Daily

UTI = Urinary Tract Infection

Note About IESU Accronym

This presentation will discuss the IESU Framework:

IESU = Indication, Efficacy, Safety, Utilization

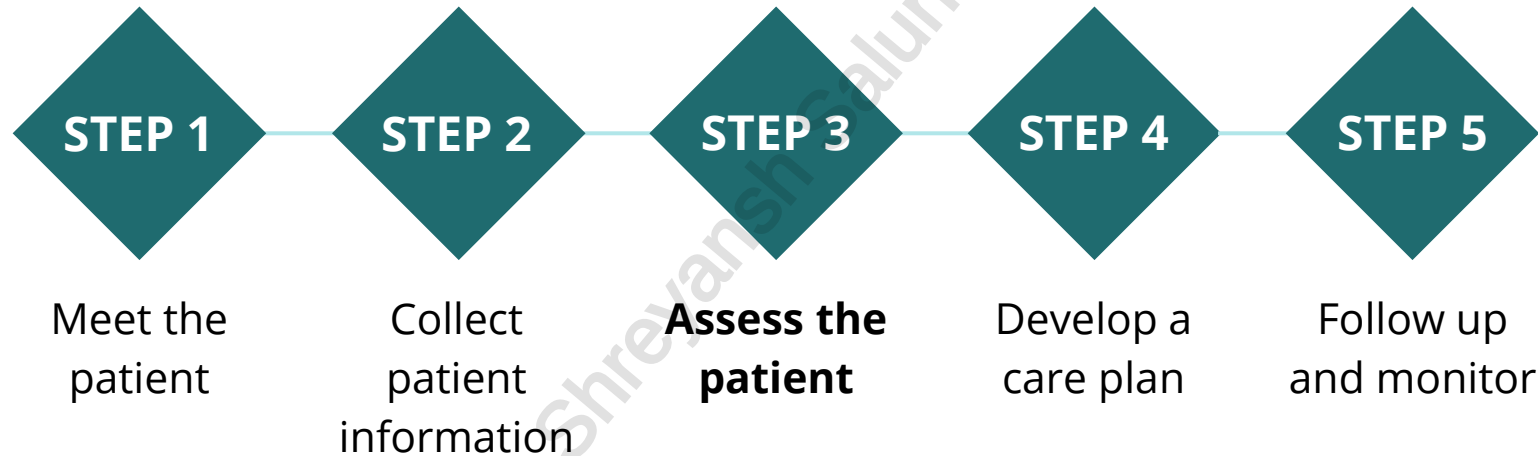
Please note that you may see **IESU** referred to as **IESA** or **NESA**

IESA = Indication, Efficacy, Safety, Adherence

NESA = Necessity, Effectiveness, Safety, Adherence

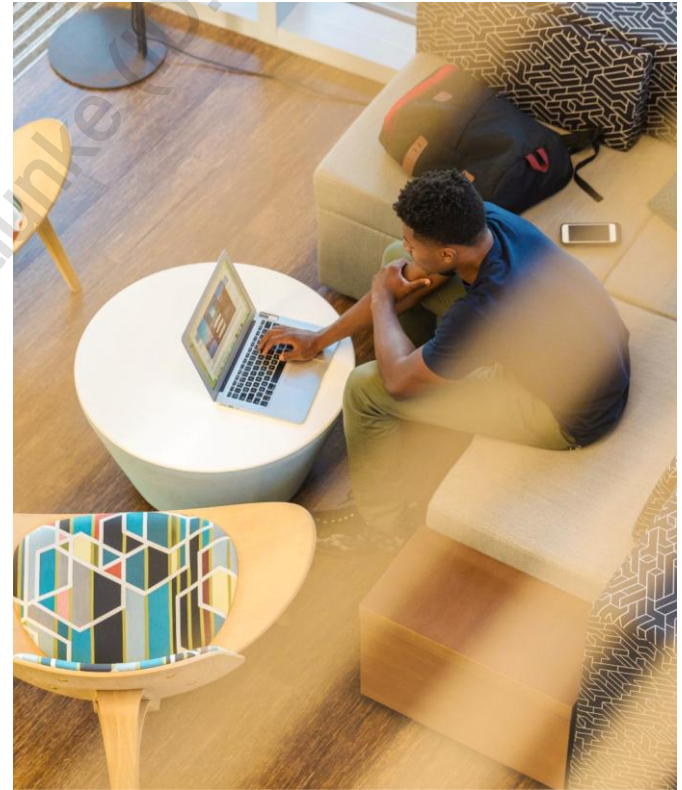
These acronyms refer to the **same framework**

Patient Care Process



Patient Assessments

- Gather relevant information from the patient and patient profile
- Use the information to identify drug-related problems (DRPs) or issues that have the potential to affect the optimization of health outcomes
- Key to having pharmacists move from a transactional practice to a more patient-centred practice





Chat, Check, Chart

Chat, Check, Chart is intended to enhance the pharmacist-patient interaction to help develop a consistent approach for evaluating medication therapy, and to promote documentation of the patient care process

Step 1

Meet the Patient

Meet the Patient

- ◇ Introduce yourself
- ◇ Confirm the patient's name and age (or date of birth)
- ◇ Obtain consent from the patient or advocate to ask questions
- ◇ Ensure confidentiality will be maintained



Meet the Patient : **Inappropriate** Patient Care Example

KL is a 26-year-old male who comes to your pharmacy with a prescription for cephalexin 500 mg PO QID x 7 days M:28 R:0

You Hello. My name is David, and I am the pharmacist on duty today. I am happy to help you with this prescription.

Hi. I need to get this prescription filled as soon as possible.

KL

You Ok, I will have this prescription filled for you. It is important to take this medication with meals and to complete the full seven days of antibiotics even if you are feeling better before the seven days are complete. Also, please note that common side effects of this medication are nausea and diarrhea.

Thank you.

KL

Meet the Patient : **Appropriate** Patient Care Example

KL is a 26-year-old male who comes to your pharmacy with a prescription for cephalexin 500 mg PO QID x 7 days M:28 R:0

You Hello. My name is David and I am the pharmacist on duty today. I am happy to help you with this prescription. How may I address you?

My name is KL. **KL**

You Nice to meet you KL. I see this prescription is written for BL. Can I confirm who BL is?

BL is my brother. The doctor wrote this prescription for him. **KL**

You How old is BL?

He is 6 years old. **KL**

You Would you know how much BL weighs?

32 kg. **KL**

You Okay, please allow me to calculate the appropriate dose of this antibiotic to ensure it is safe for your brother.

Step 2

Collect Patient Information





Used to determine the patient's knowledge and identify information needs

Includes 3 open-ended questions to gather information

Chat

	New Medication	Refill Medication
Purpose	What did your prescriber tell you this medication is for?	This medication has many uses. What do you use it for?
Direction	How did your prescriber tell you to take this medication?	How have you been taking this medication?
Monitoring	What did your prescriber tell you to expect from this medication?	Can you describe how this medication has been working for you? What side effects have you noticed?

SCHOLAR Framework

Symptoms

Characteristics

History

Onset

Location

Aggravating factors

Remitting factors



Remember to gather information about:

- Patient's health status (past or current)
- Medication history
- Allergies or intolerances
- Family history
- Social history
- Any other necessary information



SCHOLAR Framework - Examples

16

	General Questions	Examples tailored to a patient experiencing acute back pain
Symptoms	What symptoms are you experiencing?	What type of pain are you experiencing? (e.g. dull, sharp, tingling, burning)
Characteristics	Can you describe your symptoms?	Can you describe the severity of the pain on a scale of 1 to 10?
History	Have you experienced this before?	Have you had back pain before? What have you tried to relieve the pain in the past?
Onset	When did your symptoms begin?	When did your back pain start? Was there a specific event that triggered the pain?
Location	Where is the area of concern located?	Is the pain in one area or does it spread to other parts of your body?
Aggravating factors	What makes your symptoms worse?	What activities make your back pain worse?
Remitting factors	What relieves your symptoms?	What helps relieve your back pain?

BEST POSSIBLE MEDICATION HISTORY (BPMH)

17

PATIENT INFORMATION

First name: _____ Phone number: _____ Date of birth: _____ Height: _____
Last name: _____ Health ID Number: _____ Sex: _____ Weight: _____

PRIMARY HEALTHCARE PROVIDER

Full name: _____
Phone number: _____ Fax number: _____

COMMUNITY PHARMACY

Name: _____
Phone number: _____

KNOWN ALLERGIES (if applicable)

MEDICATIONS (prescription, non-prescription, natural health products)

	Medication name	Indication	Dose	Route	Frequency	Special instructions
1						
2						
3						
4						

BEST POSSIBLE MEDICATION HISTORY (BPMH)

18

PATIENT INFORMATION

First name: _____ Phone number: _____ Date of birth: _____ Height: _____
Last name: _____ Health ID Number: _____ Sex: _____ Weight: _____

PRIMARY HEALTHCARE PROVIDER

Full name: _____
Phone number: _____ Fax number: _____

COMMUNITY PHARMACY

Name: _____
Phone number: _____

KNOWN ALLERGIES (if applicable)

MEDICATIONS (prescription, non-prescription, natural health products)

	Medication name	Indication	Dose	Route	Frequency	Special instructions
1						
2						
3						
4						



Interview Guide

Collecting Patient Information: **Inappropriate** Patient Care Example

VP is a new patient to your pharmacy. She is a 78-year-old female who presents with a prescription for furosemide 20 mg PO daily. You introduce yourself as the pharmacist and confirm that the prescription is written for VP. You confirm with VP that you can ask questions and that her physician told her that she is to take this medication to help her with the fluid in her heart and to take it once a day.

You What medical conditions do you have?

I have heart failure. **VP**

You What medications do you take for your heart failure?

I am on ramipril and bisoprolol. **VP**

You Do you take any over-the-counter medications?

Just acetaminophen when I have a headache. **VP**

You Thank you for that information. I will fill this prescription for you. This medication can work to remove water from your body by causing you to pee it out. This means you may find you have to go to the washroom more often while on this medication.

Okay, thank you for this information. **VP**

Collecting Patient Information: **Appropriate** Patient Care Example

VP is a new patient to your pharmacy. She is a 78-year-old female who presents with a prescription for furosemide 20 mg PO daily. You introduce yourself as the pharmacist and confirm that the prescription is written for VP. You confirm with VP that you can ask questions and that her physician told her that she is to take this medication to help her with the fluid in her heart and to take it once a day.

You

What medical conditions do you have?

I have heart failure.

VP

You

What prescription medications do you take?

I am on ramipril and bisoprolol.

VP

You

Do you take any over-the-counter medications?

Just acetaminophen when I have a headache.

VP

You

Do you have any allergies?

Yes, I am allergic to sulfa.

VP

You

What happens if you take something containing sulfa?

The last time my throat swelled up and I could not breathe.

VP

You

Thank you for that information. Furosemide contains a sulfa component, and while the risk of an allergic reaction is rare, we should proceed with caution. I'd like to contact your doctor to discuss next steps...

Step 3

Assess the Patient



Assess the Patient

Examples of situations that require patient assessments:



A patient presenting with a new or refill prescription



A patient having their blister-pack filled



A patient asking for assistance with over-the-counter medications or a minor ailment consultation



A patient requiring a pharmacist adaptation or renewal of their prescription



A patient being recently discharged from the hospital



A prescriber calling in for a recommendation for their patient



Patient assessments are the cornerstone of providing patient-centred care and are a minimum expectation in the NAPRA Model Standards of Practice for Pharmacists

Check

Uses the 4-question IESU model to evaluate the appropriateness of therapy

I

Is the therapy **i**ndicated?

E

Is the therapy **e**ffective?

U

Is the patient willing to **u**se/adhere to the therapy?

S

Is the therapy **s**afe?



Indication

Types of Drug-Related Problems

- Unnecessary drug therapy
- An untreated medical condition requiring drug therapy

Questions to Ask Yourself

- Would non-pharmacological management be preferable to starting drug therapy at this time?
- Can drug therapy be discontinued?
- Is the recommended drug therapy a first-line option for the specific condition?
- Is an alternative drug therapy more appropriate?

Indication Examples

New Prescription	Refill Prescription
<i>Is the medication appropriate for the indication?</i> Example	<i>Is the medication still indicated?</i> Example
<i>Is the medication a first-line recommendation for the indication?</i> Example	<i>Is the medication still a first-line recommendation for the indication?</i> Example
<i>Is the medication necessary for the indication?</i> Example	<i>Is the medication still necessary for the indication?</i> Example

Indication: **Inappropriate** Patient Care Example

MB is a 62-year-old male presenting to your pharmacy with a new prescription for propranolol 10 mg PO TID M: 90 R: 3.

Click to reveal patient history

You Hello MB. Have you ever taken this medication before?

No. **MB**

You Ok, I will fill this prescription for you. I would like to inform you that propranolol is a medication used to help with headaches. Some common side effects of propranolol include low heart rate, tiredness, and dizziness. You should notice that propranolol will help with your headaches within a few weeks.

But I do not have headaches. **MB**

Indication: **Appropriate** Patient Care Example

MB is a 62-year-old male presenting to your pharmacy with a new prescription for propranolol 10 mg PO TID M: 90 R: 3.

Click to reveal patient history

You

What did your doctor tell you this medication was for?

To help control my blood pressure. MB

You

How did your doctor tell you to take this medication?

He told me to just take one pill, three times a day. MB

You

What did the doctor tell you to expect from this medication?

That it will help control my blood pressure. MB

You

Propranolol is not typically considered a first option for the treatment of blood pressure for patients in your age group. I would like to contact your physician to discuss the best option for you based on the most recent treatment guidelines for blood pressure.



Efficacy

Types of Drug-Related Problems

- Ineffective drug (e.g. incorrect drug or drug product)
- Drug therapy dose too low

Questions to Ask Yourself

- Is additional drug therapy required?
- Is the dose, frequency, or titration correct for age, weight, and organ function?
- Is the formulation correct?
- Is there potential for malabsorption issues?
- What is the onset of action?

Efficacy Examples

New Prescription	Refill Prescription
<i>Is the medication effective for the condition?</i> Example	<i>Is the medication still at an effective dose?</i> Example
<i>Is the medication at an effective dose?</i> Example	<i>Is the medication still effective for the condition?</i> Example
	<i>Is the medication working for the condition?</i> Example
<i>Is there a drug interaction preventing the medication from working?</i> Example	<i>Is there a new drug interaction that may prevent the medication from working?</i> Example

Efficacy: **Inappropriate** Patient Care Example

Ms. FG is a 64-year-old female who comes into your pharmacy to ask for a refill on her ramipril 2.5 mg PO daily. FG started on her ramipril about a year ago.

Click to reveal patient history

You

Hi FG. How are you today? Have you been experiencing any side effects from your ramipril such as cough or dizziness?

No, I am tolerating the ramipril well.

FG

You

That is great to hear. I will refill your prescription shortly and have you on your way.

Wonderful, thank you.

FG

Efficacy: **Appropriate** Patient Care Example

Ms. FG is a 64-year-old female who comes into your pharmacy to ask for a refill on her ramipril 2.5 mg PO daily. FG started on her ramipril about a year ago.

Click to reveal patient history

You

What do you take this medication for?

High blood pressure and the doctor said it is helping my heart because I have heart failure.

FG

You

How are you taking this medication?

I take it once a day in the morning.

FG

You

How have your blood pressure readings changed since starting this medication?

Well, I just checked my blood pressure on your machine twice, and it said it was 168/99.

FG

You

What side effects have you noticed since starting this medication?

None, I am tolerating this medication well.

FG

You

Based on your blood pressure readings you are reporting to me, I think it would be appropriate for you to book a follow-up appointment with your doctor to recheck your blood pressure and possibly increase the dose of your medication.



Safety

Types of Drug-Related Problems

- Adverse drug reaction
- Dose too high
- Drug interaction

Questions to Ask Yourself

- Is the patient being monitored appropriately?
- Is the dose appropriate for age, weight and organ function?
- Is the patient experiencing adverse effects? If so, can the adverse effects be managed?
- Is the patient experiencing drug-drug, drug-food, drug-disease, or drug-laboratory test interactions? If so, can the interaction be managed?

Safety Examples

New Prescription	Refill Prescription
<i>Can the medication cause harm?</i> Example	<i>Is the medication causing harm?</i> Example
<i>Is the medication at a harmful dose?</i> Example	<i>Is the patient experiencing adverse effects due to a high dose of medication?</i> Example
<i>Is monitoring required?</i> Example	<i>Has monitoring been completed recently? What were the monitoring results?</i> Example
<i>Is the patient at risk of experiencing drug interactions?</i> Example	<i>Is the patient experiencing a drug interaction?</i> Example

Safety: **Inappropriate** Patient Care Example

AL comes into your pharmacy for a refill on her levothyroxine prescription. AL has been on levothyroxine for over 10 years.

Click to reveal patient history

You Hi AL. Do you have any concerns regarding your levothyroxine?

No, I have been on it a long time. **AL**

You Ok AL. I will refill your prescription.

Thank you for your help. **AL**

Safety: **Appropriate** Patient Care Example

AL comes into your pharmacy for a refill on her levothyroxine prescription. AL has been on levothyroxine for over 10 years.

Click to reveal patient history

You What do you take this medication for?

My thyroid. **AL**

You How are you taking this medication?

I take it once a day in the morning. **AL**

You Can you describe how this medication has been working for you?

I don't really know. I haven't seen my doctor in a while, and I have not gone for a blood test in over a year. Lately, I have been feeling much more anxious than normal, and it feels like my heart is racing.

AL

You Based on the fact that you have not had a blood test in over a year and are feeling very anxious with your heart racing, I would recommend you follow up with your family doctor to have blood work done. Your symptoms may be occurring because your levothyroxine dose is too high.



Utilization

Types of Drug-Related Problems

- Overuse
- Non-adherence

Questions to Ask Yourself

- Does the patient understand the drug therapy directions (e.g. are there cultural, language or health literacy barriers)?
- Is the patient able to afford the drug therapy?
- Is the patient able to administer the drug therapy (e.g. are there age, dexterity, vision, swallowing, or memory barriers)?

Utilization Examples

New Prescription	Refill Prescription
<i>Can the patient adhere to the medication regimen?</i> Example	<i>Can the patient still adhere to the medication regimen?</i> Example
<i>Is the patient able to use the medication or medical device?</i> Example	<i>Are the side effects reducing compliance?</i> Example
	<i>Is the patient still able to use the medication or medical device?</i> Example
<i>Is the patient able to afford the medication?</i> Example	<i>Is the patient still able to afford the medication?</i> Example

Utilization: **Inappropriate** Patient Care Example

BH is a 77-year-old female who comes in with a new prescription for domperidone 5 mg PO QID before meals and bedtime M: 30 tabs R: 4. BH has a known history of type 2 diabetes and is on metformin XR 2000 mg PO daily.

[Click to reveal patient history](#)

You Hi BH, I see you have a new prescription for domperidone.

Yes, my doctor prescribed it for me. **BH**

You Ok, I will fill this prescription. Remember to take the domperidone before all meals and at bedtime. Also, please know it is typically well-tolerated but it can cause headaches.

Ok. **BH**

Utilization: **Appropriate** Patient Care Example

BH is a 77-year-old female who comes in with a new prescription for domperidone 5 mg PO QID before meals and bedtime M: 30 tabs R: 4. BH has a known history of type 2 diabetes and is on metformin XR 2000 mg PO daily.

Click to reveal patient history

- You** What did your doctor tell you this medication was for?
- I don't really remember. She just said it would help me. **BH**
- You** How did your doctor tell you to take this medication?
- She told me how to take it, but I've already forgotten what she said. **BH**
- You** What did the doctor tell you to expect from this medication?
- That it will help me. I can't remember what else she said. **BH**

- You** BH, it seems like you are having have some difficulty remembering specific details of your encounter with your doctor. Do you also find it difficult to remember to take your medications?
- Yes I do. **BH**
- You** In that case, would you like to look at options to help you remember to take your medications? I could set you up with blister packs if you think that would help you remember to take your medications.
- Yes, I would appreciate that. **BH**

Step 4 Develop a Care Plan

Developing a Care Plan

Developing a care plan occurs after using the IESU model to identify patient care issues and drug-related problems.

A patient care plan can involve:

- Referring the patient to another health care provider for follow-up and further assessment
- Referring the patient to the emergency department for urgent care
- Adapting or extending a prescription
- Sending recommendations to the prescriber or primary care provider
- Providing an immunization
- Starting a patient on compliance packaging (e.g. blister packs)
- Offering health promotion interventions (e.g. smoking cessation)
- Pharmacist initiating a prescription (e.g. prescribing for minor ailments)

Framework for Developing a Care Plan

Step	Description
Medical conditions and medication-related needs	List DRPs identified for each medical condition. Even if there is no DRP listed for a medical condition, a care plan is still necessary for monitoring
Goals of therapy	List the goals of therapy and timelines for each medical condition. Goals of therapy should be measurable or observable, and decided upon with the patient. Examples include: cure, prevent, slow progression, reduce symptoms, or normalize a lab value
Alternatives	Compare the available drug and non-drug therapies that will produce the desired goals taking into account IESU
Recommendations/Plan	In collaboration with the patient and/or other health care providers, select the best alternative and implement the plan. Here are examples of what a plan in Ontario can include: recommending drug therapy, adapting a prescription, recommending discontinuing a prescription, recommending non-drug measures, providing education, and referring a patient
Monitoring Parameters	Determine parameters to monitor for the safety and efficacy of each therapy and how the parameter is expected to change (e.g. degree of change and time frame)
Follow-up	Determine who, how, and when follow-up will occur

Examples of Adapting a Prescription vs. Contacting the Prescriber in Ontario

Adapting a Prescription

Changing the dosage form of a medication from oral tablets to a suspension for a patient who cannot swallow

Changing the dose of an antibiotic that is not dosed appropriately for a patient's weight, renal function, or indication

Changing naloxone prescribed as an injection to an easier-to-use nasal inhalation

Contacting the Prescriber

Adjusting the dose of a medication when you do not have the complete picture

Therapeutically substituting a medication



Prescribers should always be notified of any clinically significant adaptations

Medication Reviews

Ontario Professional Pharmacy Services Reference



Medication reviews occur when a pharmacist reviews a patient's prescription and non-prescription medications during an in-person or virtual interaction.

Medication reviews are useful in identifying DRPs and encourage patients to better understand their medication therapy.

Examples of when medication reviews may be considered in Ontario include:

- If a patient has just been discharged from hospital
- If a patient is switching to a blister pack
- If a patient has a chronic condition and has been on 3 or more chronic medications for their condition
- If a patient is diabetic

Steps in a Medication Review

Step 1

Schedule an appointment with the patient and take them to a comfortable place to complete the medication review or confirm a time to complete the interview over the phone (virtual)

1

2

3

Step 2

Review all of the patient's prescription and non-prescription medications using the IESU model

- Is the medication indicated?
- Is the medication effective?
- Is the medication safe or causing side effects?
- Is the patient able to use the medication?

Step 3

Document findings and provide the patient with a comprehensive drug review list that is signed and dated

Medication Review: **Inappropriate** Patient Care Example

KJ is a 62-year-old male just discharged from hospital 2 weeks ago. His current medications include metformin 1000 mg PO BID, empagliflozin 10 mg PO daily, and simvastatin 40 mg PO daily. KJ is now presenting to your pharmacy to pick up a refill on his medications.

Click to reveal patient history

You

Hello KJ. I see you were just discharged from the hospital 2 weeks ago. May I ask what happened?

I was newly diagnosed with diabetes. My blood sugar and cholesterol levels were high.

KJ

You

Thank you for that information. Do you have any questions about these new medications prescribed for you?

Not really.

KJ

You

Ok, I will have your prescriptions filled. You are on two medications for diabetes and one for your cholesterol.

Ok, thanks.

KJ

Medication Review: **Appropriate** Patient Care Example

KJ is a 62-year-old male just discharged from hospital 3 months ago. His current medications include metformin 1000 mg PO BID, empagliflozin 10 mg PO daily and simvastatin 40 mg PO daily. KJ is now presenting to your pharmacy to pick up a refill on his medications.

Click to reveal patient history

You

Hello KJ. I see you were recently discharged from the hospital. May I ask what happened?

I was diagnosed with diabetes. My blood sugar and cholesterol levels were high.

KJ

You

Thank you for that information. Would you like to sit down in this private area so I can review these medications with you?

Sure.

KJ

You

Ok, great. Metformin is a medication used to control your blood sugars. It is taken twice a day and can cause stomach upset and/or diarrhea. Can you describe any side effects you have noticed since starting metformin?

Sometimes it upsets my stomach, but I try to take it with food so then it is not as bad.

KJ

... continued on next slide

Medication Review: **Appropriate** Patient Care Example

KJ is a 62-year-old male just discharged from hospital 3 months ago. His current medications include metformin 1000 mg PO BID, empagliflozin 10 mg PO daily and simvastatin 40 mg PO daily. KJ is now presenting to your pharmacy to pick up a refill on his medications.

Patient history: You have confirmed KJ is the patient presenting to your pharmacy today, that he has no allergies, no medical conditions on file, takes vitamin D 1000 units daily but no other medication or supplements besides his 3 new prescriptions. He is a smoker and does not drink alcohol.

You

That's perfect. Empagliflozin is another medication for your diabetes, and it can increase your risk of urinary infections. Have you had any urinary infections since starting this medication?

No, but I will keep that in mind.

KJ

You

Can you describe what your blood sugar levels have been like? What about your hemoglobin A1C level?

I have not been checking my blood sugars but my A1C test from last week was 7.8%, which is down from what it was in hospital.

KJ

You

That is great. Lastly, your simvastatin is used to help with your cholesterol. Simvastatin can also cause muscle pains. Have you been noticing any muscle pains since starting on simvastatin?

Actually, I have been having really bad pain in my thighs. It is making it hard for me to walk.

KJ

You

Are you ok if I contact your doctor about your muscle pains? Due to the muscle pain you are experiencing while walking, I think you should be assessed by your doctor and your cholesterol medication may need to be changed.



Pharmacist Recommendations

Pharmacist recommendations to physicians are completed after the identification of a real or potential DRP during the course of dispensing a new or repeat prescription or when conducting a medication review.

Some examples of instances when pharmacist recommendations are warranted include:

- Duplication of medication therapy
- Subtherapeutic medication dose
- Uncontrolled medical condition
- Adverse effects of medications

Pharmacist Recommendation Example

During a medication review, you identify that KJ is having severe muscle pains, likely as a result of his newly started simvastatin 40 mg PO daily. You complete your recommendation as follows and send it to KJ's physician.



Since starting on simvastatin, KJ is experiencing significant muscle pain that is impacting his ability to walk. I would recommend having KJ's bloodwork checked to ensure there is no evidence of rhabdomyolysis, stopping his simvastatin and starting a different statin once the muscle pains resolve. I recommend rosuvastatin 5 mg daily.

Pharmacist

Patient-Centred Care Plans

- ✓ Recognize the patient's goals in the care plan
- ✓ Involve the patient in the decision-making process
- ✓ Ensure that drug-related problems are prioritized according to clinical importance as well as patient preference
- ✓ Have the patient repeat the care plan back to confirm understanding

Develop a Care Plan: **Inappropriate** Patient Care Example

JT comes to your pharmacy with a prescription for tiotropium/olodaterol 2.5/2.5 mcg 2 inhalations daily. M: 1 R: 4

Click to reveal patient history

You

Hello. How can I help you with this prescription?

I just need it filled to treat my COPD.

JT

You

Sure. May I ask JT, are you a smoker?

You

The best thing you can do for your COPD is quit smoking. I am going to sign you up for our smoking cessation program. The program will provide you with nicotine replacement therapy, as well as access to counselling and motivational interviewing to help you quit smoking.

Yes, a pack a day for 30 years now.

JT

I do not want to quit smoking.

JT

Develop a Care Plan: **Appropriate** Patient Care Example

JT comes to your pharmacy with a prescription for tiotropium/olodaterol 2.5/2.5 mcg 2 inhalations daily. M: 1 R: 4

Patient history: You have confirmed that JT is the person presenting to your pharmacy, you have consent to ask him questions, his only medical condition is COPD, he has no allergies, does not take any medications besides his new inhaler and salbutamol PRN.

You Hello. How can I help you with this prescription?

I just need it filled to treat my COPD. **JT**

You Sure. May I ask JT, are you a smoker?

Yes, a pack a day for 30 years now. **JT**

You Would you be interested in quitting smoking at this time?

Not really, but maybe in the future. I was just diagnosed with COPD and I need time to process this diagnosis before thinking about quitting smoking. **JT**

You Whenever you are ready or are looking for more information regarding quitting smoking, I am always available.

Thank you. I appreciate the information. **JT**

Step 5 Follow-up and Monitoring





Follow-up and Monitoring

- Follow-up consists of determining an appropriate timeline to assess for the efficacy and safety of a medication
- Inform the patient who follow-up will occur with (e.g. pharmacist, the physician, the laboratory to get blood work completed)
- If monitoring is required, inform the patient how often monitoring will occur
- Inform the patient of red flag signs and symptoms that should prompt them to follow up with their primary health care provider or present to the emergency room

Documentation

- A key component of follow-up and monitoring is the documentation of the patient interaction and care plan
- Documentation is required as a standard of practice, as per the NAPRA Model Standards of Practice for Pharmacists
- Documentation allows for more effective follow-up, collaboration, and continuity of care
- Documentation should be factual, complete, timely, concise, and organized
- Documentation should be readily retrievable and stored in a mutually agreed upon location



Documentation Methods

◆ SOAP

Subjective,
Objective,
Assessment,
Plan

◆ FARM

Findings,
Assessment,
Recommendations,
Monitoring

◆ DRP

Drug-related problem,
Rationale,
Plan

◆ DDAP

Drug-related problem,
Data,
Assessment,
Plan

◆ DAP

Data,
Assessment,
Plan



Documentation templates can help standardize and streamline patient assessments

Chart

Data	Patient concerns, goals, and preferences Relevant subjective and objective data about the patient Includes pertinent orders, labs, vitals
Assessment	Assessment of the drug-related problem Identification of therapeutic goals, targets, and outcomes Supporting rationale
Plan	Number the recommendations in priority or temporal sequence Drug product, dose, dosage form, route, frequency, and duration Necessary patient education or referrals Monitoring plan and follow-up

Documentation: **Inappropriate** Documentation Example

JL is asking for a refill on his atorvastatin. JL started on atorvastatin 40 mg PO daily one year ago and reports no signs of myalgia. JL has a history of CABG. JL is a known patient of yours, he is diagnosed with hypercholesterolemia, and takes atorvastatin and an omega-3 vitamin. JL has no allergies and does not smoke or drink. JL does not have any refills left on his prescription and he is leaving for a week-long vacation tomorrow, so he needs the prescription filled today. JL informs you his last blood work was done 4 months ago, and it was normal, and he is due again in 2 months. You decide to perform a pharmacist authorized renewal of JL's atorvastatin for 14 days to ensure JL has enough for his trip, and you fax his doctor to request a refill. You document the situation on the computer as follows:

JL is in need of more medication, so I refilled the prescription.

Documentation: **Appropriate** Documentation Example

JL is asking for a refill on his atorvastatin. JL started on atorvastatin 40 mg PO daily one year ago and reports no signs of myalgia. JL has a history of CABG. JL is a known patient of yours, he is diagnosed with hypercholesterolemia, and takes atorvastatin and an omega-3 vitamin. JL has no allergies and does not smoke or drink. JL does not have any refills left on his prescription and he is leaving for a week-long vacation tomorrow, so he needs the prescription filled today. JL informs you his last blood work was done 4 months ago, and it was normal, and he is due again in 2 months. You decide to perform a pharmacist authorized renewal of JL's atorvastatin for 14 days to ensure JL has enough for his trip, and you fax his doctor to request a refill. You document the situation on the computer as follows:

D: JL started on atorvastatin 40 mg one year ago after a CABG and reports no signs of myalgia. JL is leaving on a trip for 7 days tomorrow and requires a refill today, but has no refills left on his prescription.

A: Atorvastatin therapy is appropriate for cardiovascular risk reduction after a CABG. JL is adherent, tolerating atorvastatin therapy, and has had bloodwork done within the last six months.

P: Give JL 14 days worth of atorvastatin and I will follow-up with JL after his trip. I have faxed JL's doctor for future refills. I have asked JL to monitor for signs of muscle pains.

Patient Case

61



Case Example

DP is a 60-year-old male presenting to the pharmacy with a new prescription from the hospital for ciprofloxacin 500 mg BID x 7 days. DP is a known blister pack patient at your pharmacy. When you inquire about the hospital prescription, he informs you that he was admitted for a urinary tract infection (UTI) that spread to the bloodstream. DP is a retired construction worker who lives alone in his apartment. DP is a non-smoker, drinks 1 to 2 beers per day, and does not use illicit drugs. DP has an allergy to ibuprofen which causes swelling of his tongue if taken. DP's past medical history is significant for type II diabetes, hypertension, tremor, asthma, metformin-induced B12 deficiency, and osteoarthritis. DP's medications include:

- Metformin 1000 mg twice daily
- Dapagliflozin 10 mg daily
- Linagliptin 5 mg daily
- Telmisartan 80 mg daily
- Amlodipine 5 mg daily
- Propranolol 10 mg three times daily
- Salbutamol 4 puffs four times daily
- Vitamin B12 1000 mcg daily

DP is looking to fill his prescription for Ciprofloxacin. DP has his blister packs filled every Monday and has no problems using them despite his tremor. DP mentions that he has been frequently having urinary tract infections and has been noticing his tremor and breathing are worsening. DP says his breathing issues have been occurring 1 to 2 times a week for the last 2 months. DP says that his osteoarthritis has also been worse lately as he has been noticing a lot of pain in his knees. Since DP has a number of complaints, you ask him to join you in your private counselling room for a medication review.

Case Example - Continued

Laboratory Parameter	Value	Normal Lab Value
Blood Pressure	132/82 mmHg	<130/80 mmHg
Pulse	82 bpm	60 to 100 bpm
A1C	6.2%	4% to 6%
Blood glucose (fasting)	6.1 mmol/L	4 to 7 mmol/L
SCr	67 $\mu\text{mol/L}$	Men: 65 to 119 $\mu\text{mol/L}$
Vitamin B12 level	400 pmol/L	133 to 675 pmol/L

Case Example - Continued

DP Antibiotic Prescription History:

- Cephalexin 500 mg QID x 7 days 8 months ago
- Sulfamethoxazole/Trimethoprim 800 mg/160 mg Q12H x 7 days 4 months ago

Case Example - Indication

Medication	Indication
Ciprofloxacin	Bacteremia
Metformin	Type II diabetes
Dapagliflozin	Type II diabetes
Linagliptin	Type II diabetes
Telmisartan	Hypertension
Amlodipine	Hypertension
Propranolol	Tremor
Salbutamol	Asthma, not a first-line treatment option
Vitamin B12	B12 deficiency secondary to metformin
No medication	Osteoarthritis
 Statin therapy	DP should be on statin therapy based on the diabetes guidelines

Case Example - Efficacy

Medication	Efficacy
Ciprofloxacin	New medication
Metformin	Diabetes is controlled with an A1C of 6.2%
Dapagliflozin	Diabetes is controlled with an A1C of 6.2%
Linagliptin	Diabetes is controlled with an A1C of 6.2%
Telmisartan	Blood pressure is well controlled at 132/82 mmHg
Amlodipine	Blood pressure is well controlled at 132/82 mmHg
Propranolol	Tremor is worsening
Salbutamol	Breathing is worsening
Vitamin B12	Level is normal at 400 pmol/L
No medication	Osteoarthritis pain is worsening

Case Example - Safety

Medication	Safety
Ciprofloxacin	Serum creatinine is normal
Metformin	No side effects
Dapagliflozin	DP is having frequent UTIs
Linagliptin	No side effects
Telmisartan	Serum creatinine is normal, no side effects
Amlodipine	No side effects
Propranolol	Worsening breathing (drug interaction with salbutamol)
Salbutamol	High dose which can contribute to worsening tremor (drug interaction with propranolol)
Vitamin B12	No concerns

Case Example - Utilization

Medication	Utilization
Ciprofloxacin	Can add to blister pack
Metformin	Uses a blister pack that is filled regularly
Dapagliflozin	Uses a blister pack that is filled regularly
Linagliptin	Uses a blister pack that is filled regularly
Telmisartan	Uses a blister pack that is filled regularly
Amlodipine	Uses a blister pack that is filled regularly
Propranolol	Uses a blister pack that is filled regularly
Salbutamol	Has a tremor which may make it difficult to administer salbutamol appropriately
Vitamin B12	Uses a blister pack that is filled regularly

Case Example - DRPs

1. DP has been experiencing frequent urinary tract infections which could be a side effect of his dapagliflozin
2. DP is on high dose salbutamol which could be worsening his tremor
3. DP is experiencing poor control of asthma symptoms, possibly due to a drug interaction between salbutamol and propranolol, and would benefit from a maintenance medication for his asthma
4. DP is experiencing worsening of the osteoarthritis in his knees and requires referral to his primary care provider to be assessed for possible initiation of therapy
5. DP may have trouble using inhalers given his tremor
6. DP would benefit from statin therapy based on the current Diabetes Canada guidelines

Case Example - Care Plan

1. Recommend that DP's primary care provider re-assess and potentially stop DP's dapagliflozin therapy as it may be contributing to his frequent UTIs
2. Recommend that DP's physician add a low-dose inhaled corticosteroid as a controller for his asthma (such as fluticasone propionate) and change DP's salbutamol prescription to 2 puffs every four hours as needed
3. Recommend that DP's primary care provider re-assess propranolol use due to interaction with salbutamol and for tremor control
4. Assess DP's inhaler technique to ensure he is able to use his inhalers properly given his tremor
5. Refer DP to his physician for further assessment of his osteoarthritis pain
6. Recommend that DP's physician start him on a statin (such as rosuvastatin 10 mg daily)

Case Example - Monitoring and Follow-Up

Plan	Monitoring	Example of Follow-Up Timeframes
Recommend to switch ciprofloxacin	• Symptoms of UTI • Side effects of new antibiotic	• 2 to 3 days to assess improvement in symptoms • 7 days to assess resolution of symptoms
Recommend to reassess/stop dapagliflozin	• Random blood glucose • A1C • Symptoms of UTI • Symptoms of hyperglycemia	• One month to ensure no symptoms of UTI • In three months after A1C and random blood glucose are checked

Case Example - Monitoring and Follow-Up

Plan	Monitoring	Example of Follow-Up Timeframes
Start fluticasone and change salbutamol to PRN	• Symptoms of breathlessness • Number of times salbutamol is used as a rescue inhaler • Side effects of fluticasone • Reduced tremor	• 2 weeks for improvement in tremor • 4 weeks for improvement in breathing • Every 6 months to assess control of symptoms
Assess inhaler technique	• Watch DP use a metered dose inhaler • Symptoms of breathlessness due to poor inhaler technique	Reassess every 3 to 6 months, and if tremor worsens
Refer for osteoarthritis pain	• Symptoms of pain • Side effects of new medication	• Within 1 month for improvement in pain and side effects of medication • Every 3 to 6 months to reassess symptoms of pain
Start rosuvastatin	• Lipid panel • Creatine kinase • Side effects of rosuvastatin	• Within 1 month to assess side effects of rosuvastatin • 3 to 6 months to check lipid panel and creatine kinase

Case Example - Documentation

- S:** DP presented with a prescription for ciprofloxacin for his bacteremia. DP explained that he has had frequent UTIs and has been experiencing worsening of his tremor and breathing. DP also complains that his osteoarthritis has been worse lately as he has been noticing a lot of pain in his knees.
- O:** Blood pressure: 132/82 mmHg. Pulse: 82 bpm. A1C: 6.2%. Blood glucose: 6.1. SCr=67 umol/L. Vitamin B12= 400 pmol/L.
- Antibiotic Prescription History (based on pharmacy patient profile):
- Cephalexin 500 mg QID x 7 days, 8 months ago
 - Sulfamethoxazole/Trimethoprim 800 mg/160 mg Q12H x 7 days, 4 months ago

... continued on next slide

Case Example - Documentation

- A:** DP has been experiencing urinary tract infections which could be a side effect of his dapagliflozin. In addition, DP's asthma is not well-controlled with his salbutamol. He is on high dose salbutamol which could be worsening his tremors and making it more difficult for him to use his inhaler. Next, DP is experiencing worsening of the osteoarthritis in his knees but is currently not taking any medications to manage the pain. NSAIDs would be considered first-line, however DP has allergy to ibuprofen. Therefore, he should be assessed by his physician for his osteoarthritis pain. Finally, since DP is ≥ 40 years old and has diabetes mellitus, he should be on statin therapy based on the Diabetes Canada guidelines.
- P:**
1. Recommend that DP's primary care provider re-assess and potentially stop DP's dapagliflozin, since it may be contributing to his UTIs
 2. Recommend that DP's physician add a low-dose inhaled corticosteroid maintenance inhaler as a controller for his asthma (such as fluticasone propionate) and change DP's salbutamol prescription to 2 puffs Q4H PRN
 3. Recommend that DP's primary care provider re-assess propranolol due to interaction with salbutamol and for tremor control
 4. Assess DP's inhaler technique to ensure he is able to use his inhalers properly given his tremor
 5. Refer DP to his physician for further assessment of his osteoarthritis pain
 6. Recommend that DP's physician start him on a statin, such as rosuvastatin 10 mg daily, given his diagnosis of diabetes and age

Putting It All into Practice



Why Complete Patient Assessments?

- ▮ Ensures positive patient health outcomes
- ▮ Prevents medication incidents and near misses
- ▮ Establishes better patient relationships
- ▮ Increases professionalism
- ▮ Advances the profession
- ▮ Encourages a professional sense of satisfaction and fulfilment by making a difference in patient health
- ▮ Identifies opportunities for pharmacist recommendations
- ▮ Makes completing medication reviews easier as information is already available
- ▮ Meets the expectations of the profession according to the NAPRA standards of practice of pharmacists

Incorporating Patient Assessments into Practice

When implementing the patient assessments into practice, focus on one aspect of the patient assessment process at a time. For example:

- Ensuring your pharmacy layout is conducive to private patient interactions
- Making sure you as the pharmacist are easily identifiable and approachable
- Using tools or templates to help facilitate the conversation with the patient

It is important to streamline processes within the pharmacy in order to allow for more time to be spent on patient assessments. For example:

- Are the resources used for patient assessments easily accessible?
- Are there software applications available to help with patient assessments?
- Are you completing duplicate documentation? Can documentation be done electronically only?
- Is there a pharmacy technician that can support the patient assessment process?
- Is there a previous assessment you can build upon?

Tips for Incorporating Patient Assessments into Practice

Pharmacists should focus on small goals in order to successfully incorporate patient assessments into practice. Examples of small goals include:

- ◆ Focusing on a certain medical condition each month and ensuring appropriateness, effectiveness, safety, and utilization of the refill medication for that medical condition
- ◆ Completing full medication reviews and then updating key assessment information on patient records
- ◆ Setting a goal for follow-up (e.g. focusing on following up with 2 to 3 patients every dispensing shift to ensure appropriateness, effectiveness, safety, and utilization of medications)



Take Home Points

- Patient assessments are the cornerstone to providing patient-centred care
- The Chat, Check, Chart framework is an effective framework to assess new and refill prescriptions
- The patient care process consists of meeting the patient, collecting information, assessing the patient, developing a care plan, and providing monitoring and follow-up
- All prescriptions, new and refill, should be assessed for indication, efficacy, safety, and utilization
- Developing a care plan involves incorporating patient preferences and goals
- All patient assessments and interactions should be documented to allow for continuity of care

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